

Internal division of a segment



- 1 When the point P divides the $\overline{AB}$ internally, is the ratio $m : n$ positive or negative?
- 2 Point P divides the interval joining the points (x_1, y_1) and (x_2, y_2) in the ratio $m : n$.
 - a State the formula which gives the x -coordinate of point P .
 - b State the formula which gives the y -coordinate of point P .
- 3 $\overline{AB}$ is divided internally in the ratio $5 : 3$ by a point P , where A and B are the points $(1, 13)$ and $(3, 11)$ respectively.
Find the coordinates of point P .
- 4 For each of the following, find the coordinates of point P :
 - a P divides the interval joining points $(5, 10)$ and $(1, 2)$ in the ratio $3:1$.
 - b P divides the interval joining points $(2, 9)$ and $(-1, 3)$ in the ratio $2:1$.
 - c P divides the interval joining points $(-1, 10)$ and $(1, 11)$ in the ratio $4:3$.
- 5 $\triangle ABC$ has vertices at $A(4, -4)$, $B(5, 6)$ and $C(3, 4)$.
 - a Find the coordinates of P , the point which divides $\overline{AB}$ in the ratio $4 : 1$.
 - b Find the length of $\overline{CP}$.
- 6 $\triangle ABC$ has vertices at $A(3, 4)$, $B(5, 2)$ and $C(7, 7)$.
 - a Find the coordinates of point X , the point which divides $\overline{AB}$ in the ratio $5 : 1$.
 - b Find the coordinates of point Y , the point which divides $\overline{AC}$ in the ratio $5 : 1$.
 - c Find the slope of $\overline{BC}$.
 - d Find the slope of $\overline{XY}$.
 - e Is $\overline{XY}$ parallel to $\overline{BC}$?

External division of a segment

- 7 When the point P divides $\overline{AB}$ externally, is the ratio $m : n$ positive or negative?

8 For each of the following, find the coordinates of point P :



- a $\overline{AB}$ is divided externally in the ratio $-4 : 3$ by a point $P(x, y)$, where A and B are the points $(2, 9)$ and $(3, 12)$ respectively.
- b $\overline{AB}$ is divided externally in the ratio $-5 : 1$ by a point $P(x, y)$, where A and B are the points $(5, 7)$ and $(3, 1)$ respectively.

9 Point $P(x, y)$ divides the interval joining points $A(-8, 5)$ and $B(-5, 6)$ in the ratio $-3 : 2$. Find the coordinates of point P .

10 Point $P\left(1, \frac{61}{6}\right)$ divides the interval between the points $A(2, 11)$ and $B(-4, 6)$ in the ratio $m : n$.

- a What ratio does P divide $\overline{AB}$ into?
- b What ratio does B divide $\overline{AP}$ into?
- c What ratio does A divide $\overline{PB}$ into?

11 A line segment has endpoints $S(-3, 8)$ and $T(12, -4)$. Point R lies on $\overline{ST}$ such that the ratio of SR to RT is $1 : 2$. Find the coordinates of point R .